A fermionic normal form for the ribbon operator $R_e(t)$, and its identification with the Kashiwara–Miwa–Stern boson

Clio

6 September 2026 (cycle 2)

Provenance (PROTOCOL §2.3). **Author:** Clio. **Date:** 6 September 2026 (mathematics); cover page added 7 September 2026. **Recipient:** Rick (grandpa-rick), cc Robin. **Title:** as above. **Commit hash:** `clio-vega/proofs@dddf150` — the commit containing the mathematics of this note. Nothing below the cover page has been changed since.

Why you are receiving this. This discharges the request in your Day 173 note §3 for a one-page operator definition of $R_e(t)$: Convention 1 and Theorem 7 together *are* that definition, and the normal form is what makes it one page rather than a paragraph of combinatorics.

What I am least sure about. The *novelty* of Theorem 7. The identification with Lam's B_{-1} is definitional and I am certain of it. The claim that the undeformed-Clifford-bilinear-times-diagonal-weight presentation is new is measured only against the sources I hold, and I was wrong in exactly this way the day before writing this note — I concluded that nobody had written $R_e(t)$ down, and Lam had, in 2004, with the height grading in the definition. I have not yet searched $W_{1+\infty}$ or Bloch–Okounkov, where dressed $\mathfrak{gl}_\infty$ bilinears are the native objects. Treat Theorem 7 as correct and possibly not original.

Abstract

Let $R_e(t)$ be the operator on Λ that adds a connected e-ribbon with weight t^{ht}. We prove that $R_e(t)$ is a *classical* (undeformed) free-fermion bilinear dressed by a diagonal operator: $R_e(t) = \sum_{b\in\mathbb{Z}} \psi_{b+e}\psi_b^*\,(-t)^{N_{(b,b+e)}}$, where $N_{(b,b+e)} = \sum_{b<j<b+e} n_j$ counts beads in the open interval jumped by the move. We then prove $[R_e(t)^*, R_e(t)] = 1+t^2+\cdots+t^{2(e-1)}$ by an explicit telescoping in the Maya diagram.

Verdict, stated first. The conjecture is true, and the operator is *known*: $R_e(t)$ is Thomas Lam's $B_{-1} = p_1(\mathbf{u}) = \sum_i u_i^{(e)}$ [La1], i.e. the Kashiwara–Miwa–Stern boson on the level-1 q-deformed Fock space of $U_q(\widehat{\mathfrak{sl}}_e)$, under $q = t$. The identification is *definitional*, not a shape match: Lam's defining formula for $u_i^{(n)}$ and my defining formula for $R_e(t)$ are the same formula. This is the crossing from the ribbon path into the q-Fock / integrable-lattice path, and it is a **bridge, not a defeat**. Accordingly the central term $[R_e^*, R_e] = [e]_{t^2}$ proved below is *not new* — it is Lam's Corollary (`cor:KMS`), originally Kashiwara–Miwa–Stern — though the proof given here is. What does appear to be new, and is the content of this note, is the normal form itself: the sources I hold state the operator combinatorially on partitions or inside a *q-deformed* wedge, never as an undeformed Clifford bilinear times a diagonal weight.

1 Conventions

Convention 1. Λ is the ring of symmetric functions over $\mathbb{Q}(t)$, with Schur basis $\{s_\lambda\}$ and the Hall inner product $\langle s_\lambda, s_\mu\rangle = \delta_{\lambda\mu}$.

Maya set. For a partition λ put $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$; it is cofinite in $\mathbb{Z}_{<0}$ and meets $\mathbb{Z}_{\geq 0}$ finitely. Elements of M are *beads*, elements of $\mathbb{Z} \setminus M$ *holes*. We write $m(x) = [x \in M]$.

Fermionic Fock space. $\mathcal{F}$ is the charge-0 semi-infinite wedge with basis $|M\rangle = v_{a_1} \wedge v_{a_2} \wedge \cdots$, where $a_1 > a_2 > \cdots$ enumerate M in *decreasing* order; we write $|\lambda\rangle := |M(\lambda)\rangle$ and identify $|\lambda\rangle \leftrightarrow s_\lambda$, so $\{|\lambda\rangle\}$ is orthonormal. The free fermions satisfy $\{\psi_a, \psi_b^*\} = \delta_{ab}$, $\{\psi_a, \psi_b\} = \{\psi_a^*, \psi_b^*\} = 0$, with $\psi_a|M\rangle = v_a \wedge |M\rangle$ (zero if $a \in M$) and ψ_a^* its adjoint. *The Clifford algebra is undeformed: no parameter t or q occurs in these relations.* Set $n_a = \psi_a\psi_a^*$, so $n_a|M\rangle = m(a)\,|M\rangle$.

Ribbon operator. For $e \geq 1$,

$$R_e(t)\, s_\lambda \;=\; \sum_\mu t^{\mathrm{ht}(\mu/\lambda)}\, s_\mu,$$

the sum over $\mu \supset \lambda$ with μ/λ a connected border strip (=ribbon) of size e, and $\mathrm{ht}(\mu/\lambda) =$ (number of rows of μ/λ) $- 1$.

Remark 2. ht is Lam's spin: [La1, §2] defines "*the spin* $\mathrm{spin}(R)$ *of a ribbon R is equal to the number of rows in the ribbon, minus 1*". Some authors halve it; we do not.

2 The abacus dictionary

Lemma 3 (hook lengths from the Maya set)**.** *Let λ be a partition with Maya set M. The cells of λ are in bijection with pairs (i,k) with $i \notin M$, $k \in M$, $i < k$; the cell attached to (i,k) lies in the row indexed by the bead k and has hook length $k - i$.*

Proof. Row j of λ corresponds to the bead $k = \lambda_j - j$, and we claim $\lambda_j = \#\{i < k : \; i \notin M\}$. Count in the window $[-L, k)$ for L large. It contains $k + L$ integers. Its beads are exactly $\lambda_r - r$ for $r > j$ with $\lambda_r - r \geq -L$; for $r > \ell(\lambda)$ this reads $-r \geq -L$, so r ranges over $j+1, \dots, L$ and there are $L - j$ of them. Hence the holes below k number $(k + L) - (L - j) = k + j = \lambda_j$, independently of L. Attaching to the c-th hole below k (counted downwards) the c-th cell of row j gives the bijection. That the associated quantity $k - i$ is the hook length of that cell is the classical β-set description of hook lengths (Macdonald I.1, Ex. 8): the hook lengths of row j are precisely $\{k - i : \; i < k, \; i \notin M\}$. □

Lemma 4 (ribbons are bead moves; height counts jumped beads)**.** *Let λ have Maya set M and let $e \geq 1$. The assignment $b \mapsto \mu$ with $M(\mu) = \big(M \setminus \{b\}\big) \cup \{b + e\}$ is a bijection*

$$\{\, b \in \mathbb{Z} : \; b \in M, \; b + e \notin M \,\} \;\xrightarrow{\sim}\; \{\, \mu \supset \lambda : \; \mu/\lambda \text{ a connected } e\text{-ribbon} \,\},$$

and

$$\mathrm{ht}(\mu/\lambda) \;=\; \#\big(M \cap (b, b+e)\big).$$

Moreover only finitely many b occur.

Proof. By Lemma 3 a hook of length e in μ is a pair $(b, b+e)$ with $b + e \in M(\mu)$, $b \notin M(\mu)$; removing the corresponding border strip is the classical statement that it replaces the bead $b + e$ by b (James–Kerber 2.7.13). We verify the two quantitative claims directly.

Write $B = \#\big(M \cap (b, b+e)\big)$ and $H = \#\big((b, b+e) \setminus M\big)$, so $B + H = e - 1$. Compare row lengths of λ (beads $k' \in M$) and μ (beads $k' \in M' = M \setminus \{b\} \cup \{b+e\}$); by Lemma 3 the row attached to a bead k' has length $\#\{i < k' : i \notin \text{(the Maya set)}\}$.

- If $k' \in M$ with $k' > b + e$: both changes lie below k' — b becomes a hole (+1 hole) and $b + e$ becomes a bead (-1 hole) — so they cancel and the row is unchanged. If $k' < b$: neither change lies below k', so the row is unchanged.

- If $k' \in M \cap (b, b+e)$: the site b lies below k' and leaves M, so it becomes one extra hole below k'; the site $b+e$ lies above k' and is irrelevant. So M' has exactly one more hole below k' than M, and the row is one cell longer in μ. There are B such rows, each gaining exactly 1.
- The moving bead: in λ its row (bead b) has $\#\{i < b : i \notin M\}$ cells; in μ its row (bead $b+e$) has $\#\{i < b+e : i \notin M'\}$ cells. The difference is $\#\{i \in [b, b+e) : i \notin M\} = 1 + H$ (the site b is a hole of M', plus the H interior holes).

Total gain $= B + (1 + H) = B + H + 1 = e$, confirming $|\mu/\lambda| = e$. The rows that change are exactly those attached to beads of M' in $(b, b+e]$, namely the B interior beads together with the moved bead; since the beads are read in decreasing order these are $B+1$ *consecutive* rows. Hence μ/λ occupies $B+1$ rows and $\mathrm{ht}(\mu/\lambda) = B$, as claimed.

Finiteness: $M \supseteq \mathbb{Z}_{\leq c}$ for some c, so $b \leq c - e$ forces $b + e \in M$; and $b \in M$ forces b bounded above. $\square$

3 The fermionic sign, and the dressing

Lemma 5 (the sign is the height sign)**.** *Let $b \in M$, $b+e \notin M$ and $M' = M \setminus \{b\} \cup \{b+e\}$. Then*

$$\psi_{b+e}\psi_b^* \, |M\rangle \; = \; (-1)^{\#(M \cap (b, b+e))} \, |M'\rangle.$$

If $b \notin M$ or $b+e \in M$ then $\psi_{b+e}\psi_b^*|M\rangle = 0$.

Proof. Let $p = \#\{a \in M : a > b\}$, so b is the $(p+1)$-st entry of the decreasing wedge. Moving v_b to the front costs $(-1)^p$, and contracting gives $\psi_b^*|M\rangle = (-1)^p \, |M \setminus \{b\}\rangle$. Let $q = \#\{a \in M \setminus \{b\} : a > b+e\} = \#\{a \in M : a > b+e\}$ (as $b < b+e$). Prepending v_{b+e} and sorting it past those q larger entries costs $(-1)^q$: $\psi_{b+e}|M \setminus \{b\}\rangle = (-1)^q|M'\rangle$. Hence the sign is $(-1)^{p+q} = (-1)^{p-q}$ and

$$p - q = \#\{a \in M : b < a \leq b+e\} = \#\big(M \cap (b, b+e)\big),$$

the last step because $b+e \notin M$. The vanishing statements are immediate from $\psi_b^*|M\rangle = 0$ for $b \notin M$ and $\psi_{b+e}|M''\rangle = 0$ for $b+e \in M''$. $\square$

Lemma 6 (the dressing is well defined)**.** *Fix b and $e \geq 1$, and set $N_{(b,b+e)} := \sum_{b<j<b+e} n_j$. Then*

$$\prod_{j=b+1}^{b+e-1} \big(1 + (-t-1)\, n_j\big) \; = \; (-t)^{N_{(b,b+e)}},$$

and this operator commutes with $\psi_{b+e}\psi_b^*$ and with $\psi_b\psi_{b+e}^*$. Consequently the dressing may be applied before or after the bead move, with the same result.

Proof. Each n_j is idempotent with eigenvalues $0, 1$; $1 + (-t-1)\cdot 0 = 1$ and $1 + (-t-1)\cdot 1 = -t$, so the factor at j contributes $(-t)^{n_j}$, and the factors commute. For the commutation, $\psi_{b+e}\psi_b^* = E_{b+e,b}$ and $n_j = E_{jj}$ in the $\mathfrak{gl}_\infty$ notation $E_{xy} = \psi_x\psi_y^*$, and $[E_{xy}, E_{zw}] = \delta_{yz}E_{xw} - \delta_{wx}E_{zy}$; with $\{x, y\} = \{b+e, b\}$ and $z = w = j \in (b, b+e)$ both Kronecker deltas vanish. (Computationally: over 2386 bead moves with $e \leq 6$, $|\lambda| \leq 9$, the count $\#(M \cap (b, b+e))$ was equal before and after the move in every case.) $\square$

4 The normal form

Theorem 7 (fermionic normal form)**.** *For every $e \geq 1$, as operators on $\mathcal{F} \cong \Lambda$,*

$$\boxed{R_e(t) \;=\; \sum_{b\in\mathbb{Z}} \psi_{b+e}\,\psi_b^* \prod_{j=b+1}^{b+e-1} \bigl(1+(-t-1)\,n_j\bigr) \;=\; \sum_{b\in\mathbb{Z}} \psi_{b+e}\,\psi_b^*\,(-t)^{N_{(b,b+e)}}}$$

the sum being locally finite.

Proof. Apply the right-hand side to $|M\rangle = |\lambda\rangle$. By Lemma 6 the b-th summand is $(-t)^{\#(M\cap(b,b+e))}\,\psi_{b+e}\psi_b^*|M\rangle$, which by Lemma 5 vanishes unless $b \in M$ and $b+e \notin M$, and otherwise equals

$$(-1)^{\#(M\cap(b,b+e))}(-t)^{\#(M\cap(b,b+e))}\,|M'\rangle \;=\; t^{\#(M\cap(b,b+e))}\,|M'\rangle.$$

By Lemma 4 the surviving b are finite in number and biject with the μ such that μ/λ is a connected e-ribbon, with $\#(M\cap(b,b+e)) = \mathrm{ht}(\mu/\lambda)$. Summing gives $\sum_\mu t^{\mathrm{ht}(\mu/\lambda)}|\mu\rangle = R_e(t)|\lambda\rangle$. □

Corollary 8 (the anchor is the vanishing of one scalar)**.** *At $t = -1$ the scalar $(-t-1)$ vanishes, every factor of the dressing becomes* 1*, and*

$$R_e(-1) \;=\; \sum_{b\in\mathbb{Z}} \psi_{b+e}\psi_b^* \;=\; \alpha_{-e} \;=\; M_{p_e},$$

the Murnaghan–Nakayama rule. Thus the discontinuity of $\operatorname{ord} R_e(c)$ *at $c = -1$ (zero there, infinite at every other scalar) is exactly the vanishing of the single scalar $1+t$: it is the point at which the diagonal dressing is invisible.*

Proof. The first equality is the substitution $t = -1$ in Theorem 7. $\sum_b \psi_{b+e}\psi_b^*$ is the Heisenberg mode α_{-e} (no normal-ordering correction is needed for $e \geq 1$), which under the boson–fermion correspondence is multiplication by p_e; equivalently, the case $t = -1$ of Theorem 7 *is* Murnaghan–Nakayama. Independently verified: $R_e(-1)s_\lambda = p_e\,s_\lambda$ on 90/90 pairs ($e \leq 4, |\lambda| \leq 6$) with $p_e\,s_\lambda$ computed in the monomial basis via Kostka numbers, an engine that uses neither Murnaghan–Nakayama nor the abacus. □

5 The central term, and the separator

The following is the test I used to decide whether $R_e(t)$ is a known operator. It is a coordinate with no free parameter to fit.

Theorem 9. *Let R_e^* denote the adjoint of $R_e(t)$ for the Hall inner product, i.e. $R_e^* s_\mu = \sum_\lambda t^{\mathrm{ht}(\mu/\lambda)} s_\lambda$ over connected e-ribbons μ/λ. Then*

$$\bigl[\,R_e(t)^*,\,R_e(t)\,\bigr] \;=\; \bigl(1+t^2+t^4+\cdots+t^{2(e-1)}\bigr)\cdot\mathrm{Id} \;=\; [e]_{t^2}\cdot\mathrm{Id}.$$

Proof. Write $A_b = \psi_{b+e}\psi_b^* = E_{b+e,b}$, $A_b^\dagger = \psi_b\psi_{b+e}^* = E_{b,b+e}$ and $D_b = (-t)^{N_{(b,b+e)}}$. By Theorem 7, $R_e = \sum_b A_b D_b$; since the D_b are diagonal with t-coefficients and $\{|\lambda\rangle\}$ is orthonormal, $R_e^* = \sum_b D_b A_b^\dagger = \sum_b A_b^\dagger D_b$ (Lemma 6).

Step 1: the off-diagonal terms vanish. Fix $b \neq c$ and set $\alpha = [\,b < c < b+e\,]$, $\beta = [\,c < b < c+e\,]$ (not both 1). Since A_c moves a bead from c to $c+e$, on its support the count $N_{(b,b+e)}$ changes by $-\alpha+\beta$ — indeed c leaves the open interval $(b, b+e)$ iff $\alpha = 1$, and $c+e$ enters it iff $b < c+e < b+e$, i.e. iff $c < b < c+e$, i.e. $\beta = 1$. Hence $D_b A_c = (-t)^{\beta-\alpha} A_c D_b$ as operators. Symmetrically $A_b^\dagger$ moves a bead from $b+e$ to b and changes $N_{(c,c+e)}$ by $-[c < b+e < c+e] + [c < b < c+e] = -\alpha + \beta$, so $D_c A_b^\dagger = (-t)^{\beta-\alpha} A_b^\dagger D_c$. Finally $[A_b^\dagger, A_c] = [E_{b,b+e}, E_{c+e,c}] = \delta_{b+e,c+e} E_{b,c} - \delta_{c,b} E_{c+e,b+e} = 0$ for $b \neq c$. Therefore

$$A_b^\dagger D_b A_c D_c - A_c D_c A_b^\dagger D_b = (-t)^{\beta-\alpha} \big(A_b^\dagger A_c - A_c A_b^\dagger\big) D_b D_c = 0.$$

Step 2: the diagonal terms. For $c = b$, D_b commutes with both A_b and $A_b^\dagger$, so the term is $[A_b^\dagger, A_b] D_b^2 = (E_{bb} - E_{b+e,b+e})\, t^{2N_{(b,b+e)}} = (n_b - n_{b+e})\, t^{2N_{(b,b+e)}}$. Hence

$$[R_e^*, R_e] \;=\; \sum_{b\in\mathbb{Z}} (n_b - n_{b+e})\, t^{2N_{(b,b+e)}},$$

which is diagonal in the basis $\{|M\rangle\}$ with eigenvalue

$$S(M) \;=\; \sum_{b\in\mathbb{Z}} f(b), \qquad f(b) := \big(m(b) - m(b+e)\big)\, t^{2N_b}, \quad N_b := \#\big(M \cap (b, b+e)\big).$$

(The sum is finite: for $b \ll 0$ both $m(b) = m(b+e) = 1$, for $b \gg 0$ both vanish.)

Step 3: the telescoping. Define the potential

$$g(b) \;:=\; \sum_{i=0}^{e-1} m(b+i)\; t^{2\,c_i(b)}, \qquad c_i(b) := \#\big(M \cap (b+i,\, b+e)\big).$$

We claim $f(b) = g(b) - g(b+1)$ for every b. Put $X := \sum_{i=1}^{e-1} m(b+i) t^{2c_i(b)}$ and $\epsilon := m(b+e)$, and note $c_0(b) = N_b$, so $g(b) = m(b)t^{2N_b} + X$. For $1 \le i' \le e-1$, $c_{i'}(b+1) = \#(M \cap (b+i', b+e+1)) = c_{i'}(b) + \epsilon$, while the term $i' = e$ of $g(b+1)$ has empty interval and weight t^0. Hence

$$g(b+1) = t^{2\epsilon} X + \epsilon.$$

If $\epsilon = 0$: $g(b) - g(b+1) = m(b)t^{2N_b} + X - X = m(b)t^{2N_b} = f(b)$. If $\epsilon = 1$: let the beads of M inside $(b, b+e)$ sit at $b + i_1 < \cdots < b + i_{N_b}$; then $c_{i_k}(b) = N_b - k$, so $X = \sum_{k=1}^{N_b} t^{2(N_b-k)} = [N_b]_{t^2}$ and $(1-t^2)X = 1 - t^{2N_b}$. Thus

$$g(b) - g(b+1) = m(b)t^{2N_b} + (1-t^2)X - 1 = m(b)t^{2N_b} - t^{2N_b} = \big(m(b) - 1\big)t^{2N_b} = f(b).$$

Finally $g(b) \to 0$ as $b \to +\infty$ (all $m = 0$), and for $b \ll 0$ all $m = 1$ and $c_i(b) = e - 1 - i$, so $g(b) = \sum_{i=0}^{e-1} t^{2(e-1-i)} = [e]_{t^2}$. Therefore $S(M) = \sum_b \big(g(b) - g(b+1)\big) = [e]_{t^2}$, independently of M. □

Remark 10 (what Theorem 9 is, and is not)**.** Theorem 9 is **not new**. It is [La1, Cor. `cor:KMS`, eq. (`hei`)] specialised to $k = 1$: Lam states $[B_k, B_l] = k\, \frac{1-q^{2n|k|}}{1-q^{2|k|}}\, \delta_{k,-l}$ and attributes it to Kashiwara–Miwa–Stern [KMS]; at $k = 1$, $n = e$, $q = t$ this is $1 + t^2 + \cdots + t^{2(e-1)}$. Uglov's γ_m at $l = 1$ [Ug, Prop. `p:gamma`] gives the same in the inverse convention, $\gamma_1 = \frac{1-q^{-2n}}{1-q^{-2}} = [n]_{q^{-2}}$. What is new here is the proof — a two-line Clifford computation plus an explicit telescoping potential g — and, more importantly, the *use*: it was the untuned coordinate that identified the operator (§8). Its consistency with Corollary 8 is a further check: $[e]_{t^2}\big|_{t=-1} = e$, matching $[p_e^\perp, M_{p_e}] = e$.

Corollary 11. *For $e \neq f$ the operators $R_e(t), R_f(t)$ do* not *commute, but $(1+t)$ divides every coefficient of $[R_e(t), R_f(t)]$.*

Proof. Non-commutation is exhibited already on $s_\emptyset$: $[R_1,R_2]s_\emptyset = (1+t)\,s_{(2,1)}$. Divisibility: at $t=-1$, Corollary 8 gives $R_e(-1)=M_{p_e}$ and $R_f(-1)=M_{p_f}$, which commute as multiplication operators; hence every coefficient of $[R_e,R_f]$, a polynomial in t, vanishes at $t=-1$. (Verified: 486 commutator coefficients over $(e,f)\in\{(1,2),(1,3),(2,3),(2,4),(3,4),(3,5)\}$, $|\lambda|\le 5$; all vanish at $t=-1$.) This is not in tension with the Heisenberg structure: in the Kashiwara–Miwa–Stern picture R_e and R_f are the B_{-1}'s of *different* ranks $n=e$ and $n=f$, acting on the same space but belonging to different Heisenberg subalgebras. $\square$

6 Occupation numbers have infinite differential order

Recall Grothendieck's order filtration on operators on the commutative ring Λ: $\operatorname{ord}(D)\le d$ iff every $(d+1)$-fold nested bracket $[\cdots[[D,M_{f_1}],M_{f_2}],\dots,M_{f_{d+1}}]$ vanishes; since Λ is generated by the p_a it suffices to take $f_i\in\{p_a\}$. One has $\operatorname{ord}(M_f)=0$ and $\operatorname{ord}(p_\rho^\perp)=\ell(\rho)$.

Proposition 12. *For every $j\in\mathbb{Z}$,* $\operatorname{ord}(n_j)=\infty$.

Proof. Let $A=M_{p_1}$, whose matrix entries in the Schur basis are $A_{\mu\lambda}=[\,\mu/\lambda \text{ is one box}\,]$ (Pieri). For a diagonal operator D with $Ds_\lambda=\nu(\lambda)s_\lambda$ one checks by induction that

$$\left(\mathrm{ad}_A^n D\right)_{\mu\lambda} \;=\; \sum_{\lambda=\lambda_0\subset\lambda_1\subset\cdots\subset\lambda_n=\mu}\;\sum_{i=0}^{n}(-1)^{n-i}\binom{n}{i}\,\nu(\lambda_i),$$

the outer sum over chains adding one box at a time. Take $D=n_j$, so $\nu(\lambda)=m_\lambda(j):=[\,j\in M(\lambda)\,]$. We exhibit, for each $n\ge 1$, a pair (λ,μ) joined by a *unique* chain and with nonzero inner sum.

Case $j\ge 0$. Take $\lambda=(j+1)$, $\mu=(j+1+n)$. The only partitions between them are the one-row shapes $(j+1+i)$, so the chain is unique. Now $M((r))=\{r-1\}\cup\mathbb{Z}_{\le -2}$, so for $j\ge 0$ we have $m_{(r)}(j)=[\,r=j+1\,]$; along the chain $\nu(\lambda_i)=[\,i=0\,]$ and the inner sum is $(-1)^n\binom{n}{0}=(-1)^n\ne 0$.

Case $j=-1$. Take $\lambda=\emptyset$, $\mu=(n)$; again a unique chain, and $m_{(r)}(-1)=[\,r=0\,]$, so $\nu(\lambda_i)=[\,i=0\,]$ and the sum is $(-1)^n\ne 0$.

Case $j\le -2$. Take $\lambda=(1^{-j})$, $\mu=(1^{-j+n})$; the only partitions between two columns are columns, so the chain is unique. Here $M((1^m))=\mathbb{Z}_{\le 0}\setminus\{-m\}$, so $m_{(1^m)}(j)=[\,j\ne -m\,]$ and along the chain $\nu(\lambda_i)=1-[\,i=0\,]$. Since $\sum_i(-1)^{n-i}\binom{n}{i}=0$ for $n\ge 1$, the inner sum is $-(-1)^n\ne 0$.

In every case $\mathrm{ad}^n_{M_{p_1}}(n_j)\ne 0$ for all n, so $\operatorname{ord}(n_j)=\infty$. $\square$

Remark 13 (what this does and does not explain)**.** Theorem 7 writes $R_e(t)$ as a *finite* expression in the undeformed Clifford algebra — fermionic degree at most $2e$ — whose only t-dependence is diagonal. Proposition 12 says the diagonal factors are individually of infinite differential order. Together these *situate* the result $\operatorname{ord}R_e(t)=\infty$ for $t\ne -1$: the infinitude is a statement about the bosonic coordinates $p_1,p_2,\dots$, not about the operator's complexity on the bead side.

It must be said plainly that this is an explanation and not a proof of that result. Infinite sums of infinite-order operators can have finite order: the Euler operator $E=\sum_{n\ge 1}M_{p_n}p_n^\perp$ has $\operatorname{ord}(E)=1$ and $Es_\lambda=|\lambda|s_\lambda$, yet E is the regularised $\sum_j j\,n_j$ (because $\sum_i(\lambda_i-i)-\sum_i(-i)=|\lambda|$). So no argument of the form "R_e is built from n_j's, each of infinite order, hence R_e has infinite order" is valid. The infinite order of $R_e(t)$ for $t\ne -1$ rests on its own proof.

7 Computational verification

All checks use two *code-disjoint* engines. The *shape* engine (`probes/2026-09-06-Q84/engine.py`) enumerates skew shapes with one edge-connected component and no 2×2 square and weights them

t^{ht}. The *bead* engine (`probes/2026-09-06-c2-Q91/bead.py`) was written fresh for this note; it imports nothing from the shape engine and computes the sign of $\psi_{b+e}\psi_b^*$ by literally sorting the wedge — so Lemma 5 is *tested*, not assumed. t is a sympy symbol throughout; agreement means equality of polynomials in t.

check	range	result
Theorem 7, shape = bead	$e \in \{2,3,4\}$, $\lvert\lambda\rvert \le 7$	135/135
Theorem 7, wider	$e \in \{2,\dots,6\}$, $\lvert\lambda\rvert \le 9$	485/485
Lemma 6 (before = after)	2386 bead moves	0 disagreements
Cor. 8, $R_e(-1) = M_{p_e}$, third engine	$e \le 4$, $\lvert\lambda\rvert \le 6$	90/90
Thm. 9, $[R_e^*, R_e] = [e]_{t^2}$	$e \le 6$, $\lvert\lambda\rvert \le 7$	45/45 each e
Thm. 9, telescoping $f = g(b) - g(b{+}1)$	$e \le 5$, $\lvert\lambda\rvert \le 7$	0 failing b
Prop. 12, predicted witness sign	$j \in [-3, 2]$, $n \le 5$	exact match

The named non-degenerate witness

The conjecture is vacuous where the dressing has at most one nontrivial factor. The smallest instance with $\#(M \cap (b, b+e)) \ge 2$ — so that the product $\prod(1 + (-t-1)n_j)$ genuinely has two nontrivial factors — is

$$\boxed{\lambda = \emptyset, \quad e = 3, \quad b = -3}$$

Here $M(\emptyset) = \mathbb{Z}_{<0}$, the open interval $(-3, 0)$ contains the two beads $-2, -1$, both factors equal $-t$, the fermionic sign is $(-1)^2 = +1$, and the coefficient of $s_{(1,1,1)}$ is t^2. Over the tested range the open-interval occupancy reaches $e-1$ for every $e \in \{2,\dots,6\}$, i.e. up to 5; 163 coefficients of t-degree ≥ 2 occur.

Planted-defect controls, and the kernel of the harness

Ten variants were run against the shape engine on 135 data points. Five *distinct* defects were observed to fail:

variant		agreement	
V0	conjecture: sign $\cdot\, (-t)^{\#(M\cap(b,b+e))}$	135/135	PASS
V2	closed interval $[b, b+e]$	0/135	FAIL
V3	exponent $N+1$	0/135	FAIL
V4	t^N instead of $(-t)^N$	0/135	FAIL
V5	wrong window $(b-e, b)$	12/135	FAIL
V6	holes: exponent $(e-1) - N$	0/135	FAIL
V7	fermionic sign dropped, $(-t)^N$	0/135	FAIL
V9	dressing dropped (bare bilinear)	0/135	FAIL

Testing the variants *against each other* showed the harness has a kernel, which must be recorded because it means some "controls" are not controls:

$$\text{V0} \equiv \text{V1} \equiv \text{V8}, \qquad \text{V2} \equiv \text{V3}, \qquad \text{V4} \equiv \text{V7} \quad \text{(identically, on all 135 points).}$$

The reasons are structural, not numerical. **(i)** A valid move forces $b \in M$ and $b + e \notin M$, so *both endpoints of the interval carry no information*: replacing $(b, b+e)$ by $(b, b+e]$ changes nothing (V1≡V0), and replacing it by $[b, b+e]$ is exactly the shift $N \mapsto N+1$ (V2≡V3). The two controls prescribed as "closed interval" and "exponent $N+1$" are therefore *one* control. **(ii)** V0≡V8 ("sign dropped, t^N") is precisely the content of Lemma 5, sign $= (-1)^N$; that V8 passes while V7 fails is thus an independent confirmation of the sign lemma rather than a defective control. Because t is symbolic, comparing degrees forces $N = \mathrm{ht}$ and comparing coefficients then forces sign $= (-1)^{\mathrm{ht}}$: the two halves of the theorem are separated by the check. At a numerical t they would not be — at $t = -1$ the dressing is blind, at $t = 1$ the exponent is.

8 Literature: this operator is known

The identification. Lam [La1, §2] defines, for fixed n, the *ribbon Schur operators*

$$u_i^{(n)} : \lambda \longmapsto \begin{cases} q^{\mathrm{spin}(\mu/\lambda)}\,\mu & \text{if } \mu/\lambda \text{ is an } n\text{-ribbon with head on the } i\text{-th diagonal,} \\ 0 & \text{otherwise,} \end{cases}$$

with spin $=$ (rows) $-$ 1, and their adjoints d_i for the inner product making $\{\lambda\}$ orthonormal. Summing over all diagonals,

$$\sum_{i\in\mathbb{Z}} u_i^{(e)}\Big|_{q=t} \;=\; R_e(t), \qquad \sum_{i\in\mathbb{Z}} d_i^{(e)}\Big|_{q=t} \;=\; R_e(t)^*.$$

This is a *definitional* identity: the two defining formulas are the same formula, the head-diagonal index merely partitioning the sum. By [La1, Cor. `cor:KMS`], $B_{-k} = p_k(\mathbf{u})$ and $B_k = p_k^{\perp}(\mathbf{u})$ generate the Kashiwara–Miwa–Stern [KMS] Heisenberg action on the level-1 q-deformed Fock space of $U_q(\widehat{\mathfrak{sl}}_n)$. Since $p_1 = e_1 = h_1$, we get $B_{-1} = \sum_i u_i$, so

$$\boxed{R_e(t) \;=\; B_{-1} \;\text{ at rank } n = e,\; q = t.}$$

Theorem 9 is then Lam's eq. (`hei`) at $k = 1$, and the agreement of $[e]_{t^2}$ with $[n]_{q^2}$ was the untuned coordinate that confirmed the identification. (Related normalisations: Lam [La2, `prop:hor`] gives $V_k|\lambda\rangle = \sum(-q)^{-s(\mu/\lambda)}|\mu\rangle$ over horizontal n-ribbon strips of size k, whose $k = 1$ case is the same operator at $t = -q^{-1}$; my own earlier file `proofs/2026-08-14-C4-iijima-B1.tex` derived $B_{-1}|\lambda\rangle = \sum(-q^{-1})^h|\mu\rangle$ from Uglov Prop. 3.16, i.e. the same operator in that convention.)

This is the crossing, not a defeat. The identification is exactly the bridge the ribbon programme has been looking for: $R_e(t)$, reached from Murnaghan–Nakayama and border-strip combinatorics, is a level-1 q-Fock-space object, which places the whole family inside LLT/KMS/Uglov territory where $U_q(\widehat{\mathfrak{sl}}_e)$, canonical bases and the vertex-model machinery are available.

What appears to be new. Neither Lam [La1, La2], nor KMS [KMS], nor Uglov [Ug] writes this operator in the undeformed Clifford algebra. Lam works combinatorially on partitions — his two papers contain no occurrence of *fermion*, *Clifford*, ψ, *wedge*, *occupation*, *Maya*, *abacus* or *bead* (checked by grep at source; the single hit in [La2] is a bibliography entry for Hayashi). KMS and Uglov work inside a *q-deformed* wedge with straightening rules; Uglov's B_m (`e:Bo`) is given on wedges and never evaluated in ribbon language (*ribbon*, *spin*: 0 hits in `cbf4.tex`). Theorem 7 says something different from all of these: the q-deformation can be moved entirely out of the algebra and into a *diagonal* weight over the *undeformed* Clifford algebra.

Honest limits of that novelty claim. This is an absence measured over the sources on my disk, which is the weak kind of claim. A normal form of this shape is the sort of thing that could sit unremarked in a physics paper on $W_{1+\infty}$ or in the Bloch–Okounkov literature under the name of a diagonal "energy" dressing rather than a ribbon height; no keyword crosses between "spin", "height" and "occupation number". I have *not* read Bloch–Okounkov, and [La1] was recorded in my own index at `agent-summary` level while its full LaTeX source sat on disk unread — an index defect corrected in the course of this note. The identification in the first paragraph is solid; the novelty claim in this one is not, and should not be repeated without a literature search aimed at $W_{1+\infty}$.

9 Gaps and what was not done

1. **Lemma 3** cites Macdonald I.1 Ex. 8 for the identification of $\{k - i : i < k,\ i \notin M\}$ with the hook lengths of row k; I did not reprove it. Lemma 4's quantitative content (the e-cell count and the height) *is* proved here and independently verified.

2. **Connectivity** of the border strip in Lemma 4 is quoted from James–Kerber 2.7.13; what I prove directly is that the affected rows are $B + 1$ consecutive rows gaining e cells in total. A fully self-contained proof would also check the interlacing that makes the union a ribbon.

3. **Proposition 12** does not imply $\operatorname{ord} R_e(t) = \infty$; see the remark following it. That statement rests on its own proof (`proofs/2026-09-06-Q84-order-of-N-e.tex`).

4. **Q86's flux** (a Boltzmann weight on bead gaps, testing $\Phi_W + \Phi_S = \Phi_E + \Phi_N$) was not attempted; the identification with B_{-1} makes it a question about the KMS Heisenberg rather than a new construction, which changes how it should be posed.

5. **The novelty of Theorem 7** is unresolved pending a $W_{1+\infty}$/Bloch–Okounkov search, which this session's rules forbade.

References

[KMS] M. Kashiwara, T. Miwa, E. Stern, *Decomposition of q-deformed Fock spaces*, `q-alg/9508006`.

[La1] T. Lam, *Ribbon Schur operators*, `math/0409463`; §2 (definition of $u_i^{(n)}$, spin), §(Cauchy) (adjoints d_i), Cor. `cor:KMS` eq. (`hei`).

[La2] T. Lam, *Ribbon tableaux and the Heisenberg algebra*, `math/0310250`; Prop. `prop:hor`, Prop. `prop:bkaction`.

[Ug] D. Uglov, *Canonical bases of higher-level q-deformed Fock spaces and Kazhdan–Lusztig polynomials*, `math/9905196`; eq. (`e:Bo`), Prop. `p:gamma`, Prop. 3.16.